

Model Development Phase Template

Date	24 June 2026
Team ID	SWTID-2026-5981
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Feature Selection Report Template

In the forthcoming update, each feature will be accompanied by a brief description. Users will indicate whether it's selected or not, providing reasoning for their decision. This process will streamline decision-making and enhance transparency in feature selection.

Feature	Description	Selected (Yes/No)	Reasoning
CODE_GENDER	Gender of the applicant (Male/Female)	Yes	Gender can influence risk profile and income stability patterns in some scoring models.
FLAG_OWN_CAR	Whether the applicant owns a car	Yes	Ownership of assets like a car may indicate financial stability and creditworthiness.
FLAG_OWN_REALTY	Whether the applicant owns property	Yes	Real estate ownership is a strong indicator of longterm asset value and stability.

CNT_CHILDREN	Number of children the applicant has	No	Dropped to reduce noise; may not directly influence credit approval decisions in this dataset.
AMT_INCOME_TOTAL	Total annual income of the applicant	Yes	A direct indicator of the applicant's repayment ability and financial status.
NAME_INCOME_TYPE	Type of income source (e.g., Working, Pensioner)	Yes	Determines income stability and risk; key feature in risk modeling.
NAME_EDUCATION_TYPE	Highest education level	Yes	Education level often correlates with earning potential and job stability.
NAME_FAMILY_STATUS	Marital status of the applicant	Yes	Marital status can influence financial responsibilities and household income dynamics.
NAME_HOUSING_TYPE	Type of housing the applicant lives in	Yes	Indicates housing security; helps in understanding fixed living costs.
DAYS_BIRTH	Age of the applicant (converted from days)	Yes	Age influences risk—very young or old applicants may have different approval patterns.
DAYS_EMPLOYED	Duration of employment (in days, converted to years)	Yes	Longer employment generally suggests more stable income and better creditworthiness.

CNT_FAM_MEMBERS	Total number of family members	Yes	May impact disposable income and financial burden; useful for understanding repayment capacity.
OCCUPATION_TYPE	Applicant's job type	No	Dropped due to many missing values and difficulty in clean

			encoding; introduces inconsistency.
FLAG_MOBIL	Whether the mobile number is provided	No	Dropped due to constant or near-constant value across all entries; low information gain.
FLAG_WORK_PHONE	Whether the applicant provided a work phone	No	Low predictive power and correlation with target variable.
FLAG_PHONE	Whether a phone number was provided	No	Same as above; dropped to reduce noise.
FLAG_EMAIL	Whether email was provided	No_	